

Table 10: Comparison of vanilla attention with prepending a zero token and a learnable sink token during pre-training. Cache config  $x+y$  denotes adding  $x$  initial tokens with  $y$  recent tokens. Perplexity is evaluated on the first sample in the PG19 test set.

| Cache Config    | 0+1024 | 1+1023       | 2+1022 | 4+1020 |
|-----------------|--------|--------------|--------|--------|
| Vanilla         | 27.87  | 18.49        | 18.05  | 18.05  |
| + 1 Sink Token  | 1235   | <b>18.01</b> | 18.01  | 18.02  |
| + 2 Sink Tokens | 1262   | 25.73        | 18.05  | 18.05  |

as depicted in Figure 14. Our findings reveal that BERT-base-uncased exhibits the attention sink phenomenon, characterized by disproportionately high attention scores assigned to the [SEP] token in most layers. This indicates that the model consistently relies on the omnipresent [SEP] token as a focal point for attention. Furthermore, concurrent research by Darcet et al. identifies similar attention spikes in Vision Transformers, attributed to random background patch tokens acting as "registers" for global image information. We contend that these "registers" are analogous to the attention sink phenomenon we observed, suggesting that this is a universal characteristic across all Transformer models.

## I USING MORE SINK TOKENS IN THE PRE-TRAINING STAGE

Section 3.3 illustrated that incorporating a single dedicated sink token in the pre-training stage doesn't affect model performance but enhances streaming performance by centralizing attention sinks to one token. This section delves into whether adding additional sink tokens during pre-training could further optimize the performance of pre-trained language models.

As depicted in Figure 15, our experiments show that incorporating either one or two sink tokens during pre-training results in pre-training loss curves that closely resemble those of the baseline (vanilla) model. However, as detailed in Table 9, the introduction of a second sink token does not yield substantial improvements in performance across most benchmark tasks.

Further analysis, as shown in Table 10, reveals that the inclusion of additional sink tokens does not enhance streaming performance. Interestingly, the model appears to rely on both sink tokens to maintain stable streaming performance. These findings suggest that while a single sink token is adequate for improving streaming performance, adding more sink tokens does not lead to further enhancements in overall language model performance. This contrasts with findings in Vision Transformers (ViT) (Darcet et al., 2023), where multiple "registers" have been found to be beneficial.

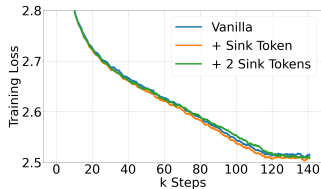


Figure 15: Pre-training loss curves of models with 0, 1, and 2 sink tokens.

Table 9: Zero-shot accuracy (in %) across 7 NLP benchmarks, including ARC-[Challenge, Easy], HellaSwag, LAMBADA, OpenbookQA, PIQA, and Winogrande.

| Methods         | ARC-c       | ARC-e       | HS          | LBD         | OBQA        | PIQA        | WG          |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Vanilla         | 18.6        | 45.2        | 29.4        | 39.6        | 16.0        | 62.2        | 50.1        |
| + 1 Sink Token  | <b>19.6</b> | <b>45.6</b> | <b>29.8</b> | <b>39.9</b> | <b>16.6</b> | 62.6        | <b>50.8</b> |
| + 2 Sink Tokens | 18.7        | 45.6        | 29.6        | 37.5        | 15.8        | <b>64.3</b> | 50.4        |